Fundamentals of AI and KR - Module 3

0. Course Introduction

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Welcome to the course

Fundamentals of Artificial Intelligence and Knowledge Representation

Module 3 – Probabilistic reasoning under uncertainty

Who I am

Roberta Calegari

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What we will cover

- Acting and reasoning under uncertainty
- Fundamentals of probability and probabilistic inference
- Independence and Bayesian network representations
- Building Bayesian networks and exact or approximate inference
- Practical examples and case studies

Contacts

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- My office: DISI Building, first floor. To gain access, dial my extension using the street-level courtesy phone.



Learning Material & Credits

Slides

A copy of these slides can be downloaded from virtuale.

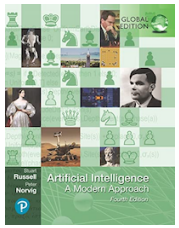
Credits

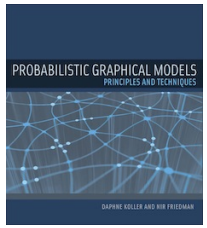
The present slides are largely an adaptation of existing material, including:

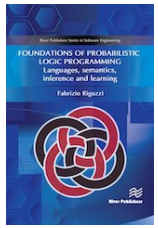
- slides from Russell & Norvig
- slides by Daphne Koller on Probabilistic Graphical Models
- slides by Fabrizio Riguzzi on Data Mining and Analytics
- slides by Paolo Torroni

I am especially grateful to these authors.

Additional Learning material







- Textbook
  - Artificial Intelligence. A Modern Approach, by Stuart Russel and Peter Norvig. Pearson Education. 2nd, 3rd Ed. Chapters 13 & 14, or 4th Ed. Chapters 12 & 13.
- Additional reading
  - Probabilistic Graphical Models. Principles and Techniques, by Daphne Koller and Nir Friedman. MIT Press.
  - Foundations of Probabilistic Logic Programming. Languages, Semantics, Inference and Learning, by Fabrizio Riguzzi. River Publishers.
  - The Book of Why: The New Science of Cause and Effect, by Judea Pearl and Dana Mackenzie. Penguin.
- Software
  - PGM Python library: pgmpy

Exam I

Exam format

Course organized into three modules

Standard assessment route

Three written exams

Module 1
WRITTEN EXAM
max 32 points
Challenges: up to +2 points, capped at 32

Module 2
WRITTEN EXAM
max 21 points
Written exam

Module 3
WRITTEN EXAM
max 11 points
Written exam

Final grade calculation

(Module 1 score + Module 2 score + Module 3 score) / 2

Round up between 18 and 30. 30 with honours only if the result is ≥ 31.0.

Special case: cross-listed course

07158 – ARTIFICIAL INTELLIGENCE
Master's Degree in Computer Science (programme code 6698)

Assessment: oral examination

PART 1 COURSE TOPICS

Oral discussion of the topics covered and explained by the students during the course

PART 2 PROJECT PRESENTATION

Students present their project

Exam II

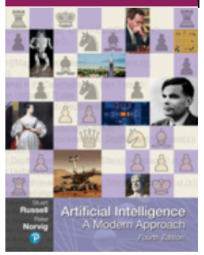
- Written exam:
  - Questions and/or exercises of the type seen in class
  - 6 dates per academic year; same day of M2
- Grades for M3 on an 11-point scale (11 is A+)
  - Other modules' max points: 32 (M1) and 21 (M2)
- Final grade is:

$$\frac{\sum_{n=1}^{3} grade(Module\ n)}{2}$$

  - Round up between 18 and 30, but 30L only if $\frac{\sum_i (grade(M_i)}{2} \geq 31.0$

Where are we in the Table of Contents?

⌂ US Edition
⌂ Global Edition
Acknowledgements
Code
Courses
Editions
Errata
Exercises
Figures
Instructors Page
Pseudocode
Reviews



Artificial Intelligence: A Modern Approach, 4th US ed.

by Stuart Russell and Peter Norvig

The authoritative, most-used AI textbook, adopted by over **1500** schools.

Table of Contents for the US Edition (or see the Global Edition)

Preface (pdf); Contents with subsections
I Artificial Intelligence
1 Introduction ... 1
2 Intelligent Agents ... 36
II Problem-solving
3 Solving Problems by Searching ... 63
4 Search in Complex Environments ... 110
5 Adversarial Search and Games ... 146
6 Constraint Satisfaction Problems ... 180
III Knowledge, reasoning, and planning
7 Logical Agents ... 208
8 First-Order Logic ... 251
9 Inference in First-Order Logic ... 280
10 Knowledge Representation ... 314
11 Automated Planning ... 344
IV Uncertain knowledge and reasoning
12 Quantifying Uncertainty ... 385
13 Probabilistic Reasoning ... 412
14 Probabilistic Reasoning over Time ... 461
15 Probabilistic Programming ... 500
16 Making Simple Decisions ... 528
17 Making Complex Decisions ... 562
18 Multiagent Decision Making ... 599
V Machine Learning
19 Learning from Examples ... 651
20 Learning Probabilistic Models ... 721
21 Deep Learning ... 750
22 Reinforcement Learning ... 789
VI Communicating, perceiving, and acting
23 Natural Language Processing ... 823
24 Deep Learning for Natural Language Processing ... 856
25 Computer Vision ... 881
26 Robotics ... 925
VII Conclusions
27 Philosophy, Ethics, and Safety of AI ... 981
28 The Future of AI ... 1012
Appendix A: Mathematical Background ... 1023
Appendix B: Notes on Languages and Algorithms ... 1030
Bibliography ... 1033 (pdf and LaTeX .bib file and bib data)
Index ... 1069 (pdf)

Exercises (website)
Figures (pdf)
Code (website); Pseudocode (pdf)
Covers: US, Global

⌂ *AI: A Modern Approach* Modified: Aug 22, 2022

https://aima.cs.berkeley.edu/

Questions?

